

Real-Time Pedestrian Detection using YOLO

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Abstract— Detecting pedestrians in a crowded scene in real time is a challenging task in monitoring and managing crowd. Many researchers around the world have addressed this task and managed to achieve satisfactory results. However, the problem of automating detection of pedestrians in the crowd is still an open issue depending on the density of crowd in a scene. To ensure safety and security, automating the crowd detection and tracking process in real time is necessary in designing a robust and secure system. Detecting and localizing objects has successfully aided in identifying the major problems with detecting pedestrians and has been a major step forward in managing crowd automatically. In this paper, we have used tiny YOLOv4. YOLO (You Only Look Once) has proved quite useful in detecting and localizing objects in an image with impressive response speed. YOLO network usually scales an entire image into fixed sized grids and then identifies and detects the region into these grids using bounding boxes. Using transfer learning on an already trained YOLO inception model on COCO dataset, detection of pedestrians in surveillance videos is handled. The paper discusses the implementation and detection performance of the proposed YOLOv4 tiny model on the UCSD pedestrian Detection dataset with promising results.

Keywords—Pedestrian Detection, YOLO, Deep Learning, COCO dataset.

I. INTRODUCTION

With abundance of CCTV footage data available, it becomes more important to accurately track and monitor crowded places. Crowd surveillance has given some very interesting problems in computer vision, especially those related to object recognition and in this case person detection, monitoring, tracking, person re-identification, and counting strength of crowds in terms of number of persons, abnormal or criminal behavior detection, mass panic detection during pilgrimage such as Kumbh Mela or Hajj-Umra etc. among others. For a few of these computer vision problems using deep learning has achieved state-of-the-art results and has performed better than humans in some scenarios. Nearly all the public places in the developed and developing countries have security cameras installed for public monitoring and safekeeping. These networks of cameras produce a huge quantity of video images for surveillance and tracking purposes. Though, monitoring all the feeds by human resources may not be an effective and feasible solution but it is reliable, and the main motive of developing Computer Vision technologies is to introduce the factor of reliability. Detection of suspicious activities or persons in video feeds can always not be accurate, even by the human experts. The imprecise nature and unexplained context of the images opens it to several interpretations whereby all or none of them may be accurate. As

context changes, the meaning of suspicious/irregular behavior changes. This opens the idea of developing techniques to describe/understand the context of the scene in question, then move further. However, a huge amount of work has already been dedicated in the Crowd Scene Analysis, such as people counting and person identification [1], etc. Classifying a suspicious/violent scene, tracking a suspicious person (Person of Interest) in a scene, re-identification using soft biometrics and occlusion in a high density crowded scene remains still unsolved to a greater extent.

The significant issue faced with Crowd Analysis using Convolutional Neural Network (CNN) is the unavailability of a functional datasets to train and test different Crowd Categories. Soft biometrics features can be used to extract and accumulate valuable information about individual persons. Categorization of people can be done using Soft biometric features and tag them while tracking them across multiple cameras. Facial recognition soft biometric features can be extracted to determine gender and ethnicity of people which can be a big step in solving the tracking problem in high density areas. The accumulated soft biometric features can be used to design a biometric feature list for each person in a surveillance video. A high-level descriptor of extracted person's soft biometric data can be entered in the database to be used for later purposes such as to track people between cameras, to build an efficient database for surveillance, to provide categorical information of people. Identifying the robust features of an individual from surveillance video and extracting them is a tedious task due to degrading quality of data and uncontrolled environment factors, such as changes in illumination, resolution, camera view angle, occlusion, and shadows which can dramatically degrade the performance of feature extraction. However, one positive side of video streamed data is the availability of multiple aspects of the facial view, and the person's snapshot at each frame as they move through the camera field of view. Some challenges such as person identification and re-identification [29-31], crowd counting [1-5] have been explored by many researchers and have been resolved to some extent but good performance still remains a challenge in obscure scenes/clips with most of the people occluded. Classification of a scene into multi-label targets such as crime scene, mob lynching, mass panic, or normal scene etc. has not been targeted up to full strength. The above-mentioned issues and challenges have been addressed to some extent with the proposed work. This work is a novel attempt to do the same.

YOLO (You Only Look Once) is a state-of-the-art object detection system that processes the image only once like a

FCNN (Fully convolutional network) and produces the output through $m \times m$ output. Unlike Faster region-based CNN (RCNN), which divides the image into various region proposals and thus performs predictions several times through each region proposal. YOLO architecture is passing the image $n \times n$ once through the FCNN architecture and splits the image into $m \times m$ grid and each grid generates bounding boxes and probabilities for these bounding boxes. YOLO frame detection is a linear regression problem instead of a complex pipeline. The image is run through the neural network to predict detections. The fastest speed of the network enables it to run on videos in real time. Unlike previous region-based techniques, that use sliding windows on the image regions to extract feature proposals, YOLO spans globally over the image while training and testing, in order to encode the contextual information about the image as well as the classes. YOLO's generalization capabilities of objects is much more accurate. When trained on natural images, and tested on simulated images, its detection performance outweighs other state of the art methods like Fast RCNN and Faster RCNN by an extended margin. YOLO Network utilizes the features from the entire image to depict the bounding boxes along with predicting the bounding boxes across all classes in the image simultaneously. This simply means that the network spans globally about the full image and about all the objects in the image.

II. RELATED WORK

In general, a crowd comprises a group of individuals where everyone has their own objectives and behavior. Practically, it is impossible to understand and analyze the behavior of everyone in a crowd. However, a collective approach can be used to study the behavior of the crowd. The following are the most recent publications in crowd surveillance.

Suspicious person detection has been studied thoroughly in recent years and has numerous applications in video surveillance [1,2,3,4,5,6,7]. While these works have achieved steady improvements over the last two decades, the increasing complexity of visual feeds is a growing obstacle that needs to be handled dynamically. These works do not make use of Deep Learning at the scale present works do, the notion of deep learning was still new and was in developmental stages. Earlier (at least 2 decades ago), it involved techniques where the feature extractor and classifier were built together, customized to suit the nature of the problem. Deep Learning techniques can extract complex and intricate structure in the images to improve the overall recognition and classification [8,9,10]. Several deep learning techniques, such as CNN [11], Recurrent Neural Network (RNN) [12], Long-Short Term Memory (LSTM) [13, 19], RCNN [14,18] and its different variants Fast-RCNN [15, 16], Faster-RCNN [17] and YOLO (You Only Look Once) [19], have been extensively used in crucial person identification and surveillance applications and shown some outstanding performances. Deep Neural Network is a compositional model, in which features are extracted ranging from low-level to high-level along the pipeline of CNN towards the external layers. The lower layers represent low-level features such as edges, and the subsequent layers represent abstract features such as shapes, etc. Deep Neural Networks have also shown significant improvements in the domain of real-time object detection [21]

where authors have trained an end-to-end classifier for joint classification, detection, and segmentation.

Authors in [20, 22, 23] have defined suspicious/anomalous events in two different categories of local and global. Local anomalies specify how the behavior of an individual differs from its neighbors whereas global anomalies are defined over the course of the entire feed. Authors in [24] have used the concept of single point and interaction-based anomaly which identifies the suspicious behavior of an individual or interaction between different entities in the scene. Importance of the context in identifying suspicious behavior was explained by [25] where activities related to each other in spatial and temporal form elaborates the context of the scene. The anomalies in the contextual form are thus identified by defining the spatial and temporal anomalies separately. Recent advances in studies on soft biometrics have utilized labels, measurements, and descriptions of individuals for identification. Soft traits possess some advantages over their traditional biometrics' counterparts. Furthermore, the attributes are human compliant, exhibit low obtrusiveness and can be inferred from far distance. Thus, soft biometrics were initially introduced as added characteristics to accompany traditional biometrics. But now the soft biometrics have explicitly proven their usefulness in re-identification of a person entering and exiting the target scene.

Soft biometrics refer physical appearance and behavioral traits of a person present in the crowd. Most recently, soft biometric based person identification has gained considerable shift because of their reliability and ease of use. In the literature, there are numerous works employing Soft-biometrics to make the system more robust in identifying people. These works have used single soft biometrics traits and integrated several of them. however, not that typical and hold several obvious limitations. Soft biometric traits naturally lack distinctiveness and permanence. A single soft biometric attribute cannot uniquely identify an individual. An intelligent combination of multiple soft attributes can produce an adequate level of identification. These limitations can be overcome by integrating several of the prominent features as shown by [26, 27]. The authors have shown with good results how well the combination of several soft biometrics traits behave in comparison with individuals and which combination is most suitable given certain conditions. Authors in [28] have extracted height and appearance to track target subjects in multi-camera setup. The most interesting experiment was shown by [29] whereby a robot is designed to identify people from a group of people based on 3 soft biometrics features in real-time. The limitation, however, is that the group is a maximum of 3 people. Clothing proved more important than height and complexion in the experiment.

GAIT analysis is another important method for identifying people in a scene. Acquiring a person's GAIT does not interfere with his/her comfort zone as it is non-invasive and does not require permission from the subject. Drunkenness, Fatigue, Variation in Clothing, Viewing Angle and carrying conditions are all the factors that limit the discriminating ability of GAIT however, the innate characteristics of a person still make it invaluable for monitoring and surveillance. GAIT is not only helpful in capturing the pose or movement of the human body

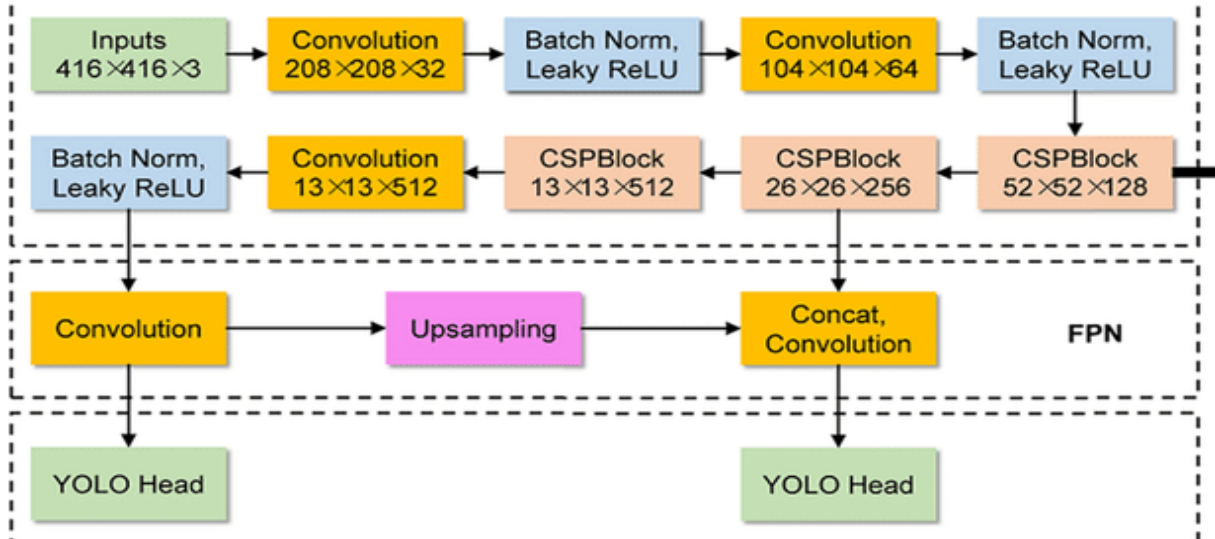


Fig. 1. YOLOv4_tiny Architecture

but also specifies certain behavioral characteristics, which is why authors in [30] have integrated GAIT information with “semantic biometrics” which is a type of soft biometrics. The score of GAIT and Semantic information are then merged by the system to identify targets in the scene. For the person re-identification task, clothes appearance has been the main cue used by [31], because of the low-quality video data being captured by most of the cameras. The authors designed SVM based on a framework for re-identification by embedding mid-level clothing attributes. Authors in [32] propose a multi-camera tracking through spatial and appearance features. The position of the target object is approximated using spatial features and color is used for appearance feature matching. A soft Biometrics based system using hair, gender, backpack, and clothing is proposed in [33] where features are detected, and regression is used to calculate attribute-based distance between pairs of images for each feature. Around 70% of the pedestrians captured in street scenes are occluded in at least one video frame. At present, most of the existing vision-based solutions to extract low-level features of crowds belong to the object-based genre which often begins with analyzing crowd behaviors through the segmentation, tracking and detection of individuals within successive image frames.

III. METHODOLOGY

In this work, we have experimented with a pre-trained machine learning model for performing pedestrian detection/people detection from a video feed. This video feed can either be input from a webcam or a stored video file in the system. The pretrained model we fine-tuned for experiments is YOLO4-tiny. The experiment does not involve training a model from scratch as pre-trained models of such kind are freely available for different academic and research purposes rather, they are fine-tuned for a new dataset. YOLO4 -tiny is the fourth available version of the reputed YOLO (You Only Look Once) model which was developed by Joseph Redmon as an advanced

real-time object detection model. The YOLO4 architecture comprises CSPDarknet53 as a backbone, spatial pyramid pooling additional module, PANNet-aggregation neck and YOLOv3 Head. The CSPDarknet53 is a convolutional foundation that enhances the learning and predicting capability of the underlying CNN module. The major difference between YOLOv4 and YOLOv4 tiny is the significant reduction of network size. The number of convolutional layers in the backbone of the architecture are compressed. It can trim 91% of the standard YOLOv4 model size by compressing 23 CSPDarknet layers backbone into three CSPDarknet layers and using only two scale detectors. The figure 1 shows the architecture of the YOLOv4-tiny model that we have used in our experiments.

IV. RESULTS & DISCUSSION

The required experiments are performed using Python 3.10 on a NVIDIA GPU RTX2070 with MAX-Q design. The fine-tuning of hyperparameters is done with little or no cost incurred. Minimum confidence threshold is set to 0.2 for the score returned by the model above which a prediction is supposed to be true. The model takes the input image frame by frame in form of dimension and is passed to the YOLO model which then performs a forward pass and returns a bounding box for the detections in the frames and confidence scores for each detection. From the individual detection we get the scores of all the predictions, class id and confidence for the class id with maximum score. The class id for the person is ‘O’ and its confidence is above threshold in order to avoid false positives. To remove overlapping and weak bounding boxes, non-maxima suppression is performed because neighboring windows of actual detection also contain high scores, hundreds of predictions need to be cleaned up. The Designed model was tested on UCSD Pedestrian Dataset. The dataset contains videos of pedestrians standing, walking away, parallel to the camera.

This section summarizes the various performance metrics of the predictions made on the UCSD Pedestrian Dataset. The three test sets composed were a combination of samples randomly chosen from the entire test set. The first two sets were mutually exclusive and there was no repetition of samples in them. The third test set was designed with replacement sampling in order to test the performance of the system on similar type of input occurring multiple times. The test performance of the proposed method was evaluated in parts on a test set of UCSD pedestrian Dataset.

In order to evaluate the performance of the proposed method, The predicted bounding boxes of pedestrians are compared to the ground truth bounding boxes. Following metrics have been used to showcase the performance of the proposed method and to compare with the previous works.

- Accuracy is the ratio of correct predictions to all predictions. It reflects how well the model can classify a region as pedestrian or not.
- F1 score is a weighted average of precision and recall. It reflects how well the model can balance between precision and recall.
- mAP (mean Average Precision) average of the precision values for different recall levels. mAP reflects how well the model can detect pedestrians at various confidence thresholds. These results metrics have widely been used by researchers for the object detection and localization performance evaluation for deep learning systems.

TABLE I. SUMMARY OF THE RESULT METRICS OF PROPOSED MODEL ON DIFFERENT TEST SETS

S.No.	TPR	FPR	F-score	AUC	mAP
Test dataset1	0.886	0.113	0.7943	0.9082	0.78
Test dataset2	0.915	0.085	0.8187	0.9361	0.82
Test dataset3	0.903	0.007	0.8039	0.9228	0.85

The AUC score of the 3test dataset was observed above 90% which is comparable to the results obtained on the same dataset using other state of the art methods. These results can further be improved by deploying some proven state of the art pre-processing techniques that could further enhance the detection power at the cost of time and speed.

V. COMPARATIVE ANALYSIS

In this section we compared the performance of three versions of YOLO: YOLOv2 [34], YOLOv3 [35] and YOLOv4 [36] using the same UCSD pedestrian Dataset. Some of the key factors that contribute to the superior performance of YOLOv4 are: (1) the use of a modified CSPDarknet53 backbone with SPP and PANet modules, which enhances the feature extraction and fusion capabilities of the network; (2) the use of a novel SAM (Spatial Attention Module) block, which improves the feature selection and localization accuracy; (3) the use of a combination of data augmentation techniques, such as CutMix, Mosaic and Random Affine Transformation, which increase the diversity and robustness of the training data; which further improves the generalization of the model, achieving high accuracy and speed. The comparison is summarized in Table II.

TABLE II. COMPARISONS OF YOLO VERSIONS WITH PROPOSED WORK

Models	Accuracy (%)	F1-score	mAP
YOLO v2*	86	0.71	0.63
YOLO v3*	89	0.76	0.69
YOLO v4 (tiny) (proposed work)	91	0.79	0.73

The models marked by asterisk () were not implemented by the authors rather taken from literature. The comparison performance with Test dataset3 is not available as existing works were not tested for this dataset.

As we can see, the proposed & implemented YOLOv4 model achieves the highest accuracy among the three models, with a significant improvement over YOLOv2 and a slight improvement over YOLOv3. YOLOv2 has the lowest accuracy.



Fig. 2 The resultant detection performance of the proposed work on the randomly captured dataset on the campus of DCS, JMI, New Dehi-25

The v4 and v4-tiny are considered as the same models for test result comparisons.

VI. CONCLUSION

The discussed method uses a pre-trained YOLOv4-tiny model to design a real time robust and much accurate pedestrian detection system. However, it is a pretrained model and we only fine-tuned the model during its training phase by using transfer learning. One of the major challenges in pedestrian detection is handling occlusion in real time scenarios. This pretrained YOLOv4-tiny model can further be fine-tuned by training on an occluded dataset and thus enhancing prediction and detection

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